

The z integration, done first

Every step, and the final result for the $\mathfrak{A}^{(1)\dagger}\bar{\mathfrak{C}}^\dagger \times \mathfrak{B}_2$ row

$$S = p^+ + k^+, \quad \beta = \frac{p^+}{k^+}, \quad B = (x - w)^2, \quad X^{k'} = \frac{(x' - y')^{k'}}{(x' - y')^2}, \quad u = z - y', \quad \hat{k} = k_\perp / |k_\perp|$$

First, the honest constraint: this z integral **cannot** be done in closed form, because the colour factor contains $U(z)$ — a Wilson line at the integration point. No amount of algebra removes it; it depends on the target field.

What can be done, and is the standard and correct thing: split the integral into an *asymptotic piece that carries all the logarithms and is exactly integrable*, plus an *absolutely convergent remainder*. That is what §3–§6 do. The result:

$$\int d^2 z \mathcal{I}(z) = \underbrace{\int d^2 u \left[\mathcal{I} - \theta(|u| - \rho_0) \mathcal{I}_{\text{asy}} \right]}_{\text{absolutely convergent, } |u|^{-3} \text{ tail}} + \underbrace{e^{-ik \cdot (y' - w)} \mathcal{C}_\infty \frac{\pi}{p^+} \left\{ \frac{2\mathcal{K}}{k^+} L + \frac{p^{+2}}{k^{+2} S} \left[(L + \tfrac{1}{2}) \mathcal{K} - \mathcal{K}_k \right] \right\}}_{\text{exact}}$$

with $L = \log \frac{2e^{-\gamma_E} k^+}{p^+ |k_\perp| \rho_0}$, and the p^+ integral of the second piece is elementary (§8), giving the $\log^2$ in closed form. Both halves verified numerically (§7).

A correction to what I told you last time: the coefficient of the double log is $\frac{2\pi\mathcal{K}}{k^+}$, not the -0.0490 I quoted. That number came from a least-squares fit over too wide a p^+ range; the correct value is derived analytically here and confirmed by the converging local slope (§7.3). The *existence* of the $\log^2$ stands.

0. Why there is no closed form, and what to do instead

After the w' integration the integrand is

$$\mathcal{I}(z) = e^{-ik \cdot (y' - w)} e^{-i\beta k \cdot (y' - z)} \frac{N^m X^{k'}}{D} \mathcal{R}^{mk'}(z) \mathcal{C}(z), \quad N^m = \frac{(y' - z)^m}{(y' - z)^2}, \quad D = p^+(x - z)^2 + k^+(x - w)^2$$

and z appears in **five** places: N^m , the phase, D , the bracket $\mathcal{R}^{mk'}$, and the colour factor $\mathcal{C}(z)$ through $U(z)$. The first four are ordinary functions and could in principle be integrated. $U(z)$ cannot: it is a path-ordered exponential of the target field at the integration point.

So the goal is not "do the integral" — it is "extract everything that is divergent or logarithmically enhanced in closed form, and leave behind something that converges absolutely and can be evaluated once you specify the target." That is a subtraction, and it is how every calculation of this type is organised. The piece you extract must (i) reproduce the exact large- $|z|$ behaviour, and (ii) be integrable in closed form.

1. Change of variable: use $u = z - y'$, not z

This looks cosmetic and is not. Two reasons:

- **Exactness.** $N^m = \frac{(y' - z)^m}{(y' - z)^2} = -\frac{u^m}{u^2}$ — exactly, not just asymptotically.
- **The phase.** $e^{-i\beta k \cdot (y' - z)} = e^{+i\beta k \cdot u}$. If instead you centre on z , you must write $e^{-i\beta k \cdot y'} e^{+i\beta k \cdot z}$, and the stray factor $e^{-i\beta k \cdot y'}$ sits *outside* the z integral while still depending on $\beta = p^+/k^+$ — it would then contaminate the p^+ integral and destroy the clean $\log^2$. Centring on y' puts **all** the p^+ -dependent phase inside the transverse integral, where the Bessel masters absorb it.

The measure is unchanged: $d^2 z = d^2 u$ (a shift, Jacobian 1).

2. Power counting: what survives at large $|u|$

We need the integrand to order $|u|^{-2}$, since $\int d^2 u |u|^{-2}$ is the log. Collect the scalings ($A = (x - z)^2 \rightarrow |u|^2$, $C = (z - w)^2 \rightarrow |u|^2$, B fixed):

$$N^m = -\frac{u^m}{u^2} \sim \frac{1}{|u|}, \quad \frac{1}{D} \rightarrow \frac{1}{p^+ |u|^2}, \quad X^{k'} \text{ fixed}$$

so we need the $O(|u|)$ part of $\mathcal{R}^{mk'}$. Going through the six groups with $\mathcal{V}_w \rightarrow -\frac{x-w}{2B}$ (order 1) and $\mathcal{V}_z \rightarrow \frac{u}{2|u|^2}$ (order $|u|^{-1}$):

group	structure	size	keep?
$d_\perp$	$\delta^{mk'} \frac{p^+}{2CS^2} (A - B) \rightarrow \frac{d_\perp p^+}{2S^2}$	$O(1)$	no
S/k^+	$\delta^{mk'} \frac{p^+}{k^+ S} \left[\dots - \frac{(x-z) \cdot (x-w)}{2B} \dots \right] \rightarrow + \frac{p^+}{k^+ S} \frac{u \cdot (x-w)}{2B}$	$O(u)$	yes
S/p^+	all three terms $O(1)$	$O(1)$	no
$-1/p^+$	$\mathcal{V}_z^m (x-w)^{k'}$	$O(u ^{-1})$	no
$-p^+/k^{+2}$	$\mathcal{V}_w^m (x-z)^{k'} \rightarrow \left(-\frac{(x-w)^m}{2B} \right) (-u^{k'})$	$O(u)$	yes
$-1/k^+$	$(x-z)^m \mathcal{V}_w^{k'} \rightarrow (-u^m) \left(-\frac{(x-w)^{k'}}{2B} \right)$	$O(u)$	yes

(Note it is the *growing* factors $(x-z)^m \rightarrow -u^m$ that promote three groups to $O(|u|)$; the soft-enhanced $-1/p^+$ group is **not** among them.)

Keeping only those three:

$$\mathcal{R}^{mk'} \longrightarrow \delta^{mk'} \frac{p^+}{k^+ S} \frac{u \cdot (x-w)}{2B} - \frac{p^+}{k^{+2}} \frac{(x-w)^m u^{k'}}{2B} - \frac{1}{k^+} \frac{u^m (x-w)^{k'}}{2B}$$

Contract with $N^m X^{k'} = -\frac{u^m}{u^2} X^{k'}$ and divide by $D \rightarrow p^+ |u|^2$:

- group S/k^+ : $-\frac{p^+}{k^+ S} \frac{(u \cdot X)(u \cdot (x-w))}{2B |u|^2}$
- group $-p^+/k^{+2}$: $+\frac{p^+}{k^{+2}} \frac{(u \cdot (x-w))(u \cdot X)}{2B |u|^2}$
- group $-1/k^+$: $+\frac{1}{k^+} \frac{(x-w) \cdot X}{2B}$ (using $u \cdot N = -1$)

The first two combine, $\frac{p^+}{k^{+2}} - \frac{p^+}{k^+ S} = \frac{p^+}{k^+} \frac{S - k^+}{k^+ S} = \frac{p^{+2}}{k^{+2} S}$:

$$\mathcal{I}_{\text{asy}}(u) = e^{-ik \cdot (y' - w)} \mathcal{C}_\infty \frac{e^{i\beta k \cdot u}}{|u|^2} \frac{1}{p^+} \left[\frac{\mathcal{K}}{k^+} + \frac{p^{+2}}{k^{+2} S} \frac{(x-w)^m X^{k'}}{2B} \hat{u}^m \hat{u}^{k'} \right]$$

$$\mathcal{K} \equiv \frac{(x-w) \cdot X}{2B} = \frac{(x-w) \cdot (x' - y')}{2(x-w)^2 (x' - y')^2}$$

Checked: the angular average of $\mathcal{I}$ times $|u|^2$ against $\frac{\mathcal{K}}{p^+} \left[\frac{1}{k^+} + \frac{p^{+2}}{2k^{+2} S} \right]$ at $p^+ = 0.05, 0.15, 0.37, 1.0, 3.0$: ratio = 1.00000 to five decimals in every case.

3. The colour at large $|u|$

Far from the target $U(z) \rightarrow 1$. Setting $U(z) = 1$ in the (uniformly ordered) colour factor:

$$\text{bra} \rightarrow f^{c'ba} [U(y') - U(x')]^{c'e'} \rho^{e'}(x'), \quad \text{ket} \rightarrow f^{cbd} U^{ac}(w) \rho^d(x) - f^{abd} U^{de}(x) \rho^e(x)$$

The second product collapses: $-\sum_{a,b} f^{c'ba} f^{abd} = +\sum_{a,b} f^{c'ab} f^{abd} = +N_c \delta^{c'd}$ (verified to $2 \cdot 10^{-15}$). The first does not — $\sum_b f^{c'ba} f^{cbd}$ is an irreducible four-index object. So

$$\mathcal{C}_\infty = [U(y') - U(x')]^{c'e'} \rho^{e'}(x') \left\{ f^{c'ba} f^{cbd} U^{ac}(w) \rho^d(x) + N_c U^{c'e}(x) \rho^e(x) \right\}$$

a constant (in u), which therefore comes outside the transverse master integrals.

4. The two Bessel masters

4.1 The scalar one

Average over the direction of u first, using $\langle e^{iq \cdot u} \rangle_{\hat{u}} = J_0(q|u|)$:

$$\mathcal{M}_0(q, \rho_0) \equiv \int_{|u| > \rho_0} d^2 u \frac{e^{iq \cdot u}}{|u|^2} = 2\pi \int_{\rho_0}^{\infty} \frac{dr}{r} J_0(qr) = 2\pi \int_{q\rho_0}^{\infty} \frac{J_0(t)}{t} dt$$

and for $q\rho_0 \ll 1$ the standard expansion (from $\int_0^\infty J_0(t) t^{s-1} dt = 2^{s-1} \Gamma(\frac{s}{2}) / \Gamma(1 - \frac{s}{2})$ near $s = 0$) gives

$$\mathcal{M}_0 = 2\pi \left[\log \frac{2}{q\rho_0} - \gamma_E \right] + O((q\rho_0)^2)$$

Verified with oscillatory quadrature: at $q\rho_0 = 10^{-2}, 10^{-4}, 10^{-6}$ the numerical values match to $1.2 \cdot 10^{-5}, 1.2 \cdot 10^{-9}, 1.2 \cdot 10^{-13}$ — i.e. the error really is $O((q\rho_0)^2)$. (Plain quadrature gets this wrong by $5 \cdot 10^{-3}$; the tail must be done with an oscillatory method.)

4.2 The tensor one

Using the exact two-dimensional angular identity

$$\langle \hat{u}^m \hat{u}^{k'} e^{it \hat{u} \cdot \hat{q}} \rangle_{\hat{u}} = \frac{J_1(t)}{t} \delta^{mk'} + \left(J_0(t) - \frac{2J_1(t)}{t} \right) \hat{q}^m \hat{q}^{k'}$$

(check by taking the trace: $\frac{2J_1}{t} + J_0 - \frac{2J_1}{t} = J_0$ ✓), one gets

$$\begin{aligned} \mathcal{M}_2^{mk'} &\equiv \int_{|u| > \rho_0} d^2 u \frac{\hat{u}^m \hat{u}^{k'}}{|u|^2} e^{iq \cdot u} = 2\pi \left[a_1 \delta^{mk'} + a_2 \hat{q}^m \hat{q}^{k'} \right] \\ a_1 &= \int_{q\rho_0}^{\infty} \frac{J_1(t)}{t^2} dt, \quad a_2 = \int_0^{\infty} \left(J_0(t) - \frac{2J_1(t)}{t} \right) \frac{dt}{t} \end{aligned}$$

a_2 is **finite** at small t (the integrand is $\simeq -t/8$), and numerically it is exactly

$$a_2 = -\frac{1}{2} \quad (\text{to all 20 digits computed})$$

while a_1 follows from the trace constraint $2a_1 + a_2 = \mathcal{M}_0/2\pi$, i.e. $a_1 = \frac{1}{2} \left[\log \frac{2}{q\rho_0} - \gamma_E + \frac{1}{2} \right]$ (verified to $3 \cdot 10^{-14}$). Hence

$$\mathcal{M}_2^{mk'} = \pi \left[\left(L + \frac{1}{2} \right) \delta^{mk'} - \hat{q}^m \hat{q}^{k'} \right], \quad \mathcal{M}_0 = 2\pi L, \quad L \equiv \log \frac{2}{q\rho_0} - \gamma_E$$

Note the shape: the same $[(\log) \delta - \hat{q} \hat{q}]$ structure as the $T^{mm'}$ master of the two- $\mathfrak{C}$ row, and the log sits in the δ part with half weight — which is just the angular average $\langle \hat{u}^m \hat{u}^{k'} \rangle = \frac{1}{2} \delta^{mk'}$.

5. Assembling the asymptotic piece

Here $q = \beta k_\perp$, so $|q| = \frac{p^+ |k_\perp|}{k^+}$ and $\hat{q} = \hat{k}$, giving

$$L = \log \frac{2e^{-\gamma_E} k^+}{p^+ |k_\perp| \rho_0} \equiv \log \frac{\Xi}{p^+}, \quad \Xi \equiv \frac{2e^{-\gamma_E} k^+}{|k_\perp| \rho_0}$$

Contracting $\frac{(x-w)^m X^{k'}}{2B}$ with $\mathcal{M}_2^{mk'}$ gives $\pi \left[\left(L + \frac{1}{2} \right) \mathcal{K} - \mathcal{K}_k \right]$ with $\mathcal{K}_k \equiv \frac{[(x-w) \cdot \hat{k}][X \cdot \hat{k}]}{2B}$, and $\frac{\mathcal{K}}{k^+} \mathcal{M}_0 = \frac{2\pi \mathcal{K} L}{k^+}$. Therefore

$$\mathcal{G}_{\log}(p^+) \equiv \int_{|u| > \rho_0} d^2 u \mathcal{I}_{\text{asy}} = e^{-ik \cdot (y' - w)} \mathcal{C}_\infty \frac{\pi}{p^+} \left\{ \frac{2\mathcal{K}}{k^+} L + \frac{p^{+2}}{k^{+2} S} \left[\left(L + \frac{1}{2} \right) \mathcal{K} - \mathcal{K}_k \right] \right\}$$

6. The subtraction

$$\int d^2 z \mathcal{I} = \underbrace{\int d^2 u \left[\mathcal{I}(u) - \theta(|u| - \rho_0) \mathcal{I}_{\text{asy}}(u) \right]}_{\mathcal{G}_{\text{fin}}(p^+)} + \mathcal{G}_{\log}(p^+)$$

ρ_0 is arbitrary: any change in it moves the same amount between the two pieces. Choose it of order the typical transverse separation so that $\mathcal{G}_{\text{fin}}$ is small.

7. Numerical verification

7.1 The remainder converges absolutely

$ u $	$\langle \mathcal{I} \rangle u ^2$	$\langle \mathcal{I} - \mathcal{I}_{\text{asy}} \rangle u ^3$
10^3	0.0101234	0.264
10^4	0.0101241	2.882
10^5	0.0101241	4.402

$\mathcal{I}$ itself has the flat $|u|^{-2}$ tail (middle column). The remainder does not: multiplied by $|u|^3$ it grows by only a factor 1.5 per decade, not 10, so it falls faster than $|u|^{-2}$ and $\int d^2u$ of it converges absolutely. (It is between $|u|^{-3}$ and $|u|^{-3} \log |u|$; either is fine.)

7.2 The closed form is right where it matters

p^+	$q\rho_0$	numeric	closed form	rel. diff
0.300	0.208	$0.012947 - 0.004384i$	$0.014048 - 0.004757i$	$7.8 \cdot 10^{-2}$
0.100	0.070	$-0.539560 + 0.182705i$	$-0.539436 + 0.182663i$	$2.3 \cdot 10^{-4}$
0.030	0.021	$-2.805504 + 0.949997i$	$-2.805555 + 0.950014i$	$1.8 \cdot 10^{-5}$
0.003	0.002	$-44.510354 + 15.072047i$	$-44.511279 + 15.072360i$	$2.1 \cdot 10^{-5}$

The error is $O((q\rho_0)^2)$ and disappears exactly in the small- p^+ region where the logarithms are generated. If you want the formula at large p^+ as well, keep $\mathcal{M}_0 = 2\pi \int_{q\rho_0}^{\infty} \frac{J_0(t)}{t} dt$ unexpanded — that form is exact.

7.3 The decomposition reassembles, and fixes my earlier coefficient

p^+	remainder	+ closed form	= total	direct
0.10	0.262443	-0.539436	-0.276993	-0.277910
0.03	-0.485798	-2.805555	-3.291352	-3.288024
0.01	-3.157916	-10.792209	-13.950126	-13.941670

Agreement to 0.03%–0.06%, which is the quadrature accuracy.

Correction to the previous note. I quoted the double-log coefficient as -0.0490 from a least-squares fit. That fit used points up to $p^+ = 0.1$, where the asymptotics has not set in. The analytic coefficient derived above is $\frac{2\pi K}{k^+} = -0.022416$, and the *local* slope of $I(p^+)p^+$ against $\log \frac{k^+}{p^+}$ converges to it:

p^+	$3 \cdot 10^{-2}$	10^{-2}	$3 \cdot 10^{-3}$	10^{-3}	$3 \cdot 10^{-4}$	10^{-4}	analytic
local slope	-0.0639	-0.0365	-0.0283	-0.0245	-0.0233	-0.0226	-0.02242

The **existence** of the $\log^2$ is unchanged; only its coefficient is corrected, and it is now derived rather than fitted.

8. The p^+ integral, in closed form

8.1 The leading (double-log) term

$$\int_{\Lambda}^{\vee-k^+} \frac{dp^+}{p^+} \log \frac{\Xi}{p^+} = \left[-\frac{1}{2} \log^2 \frac{\Xi}{p^+} \right]_{\Lambda}^{\vee-k^+} = \frac{1}{2} \left[\log^2 \frac{\Xi}{\Lambda} - \log^2 \frac{\Xi}{\vee-k^+} \right]$$

(primitive verified symbolically: $\frac{d}{dp} \left[-\frac{1}{2} \log^2 \frac{\Xi}{p} \right] - \frac{1}{p} \log \frac{\Xi}{p} = 0$.)

8.2 The subleading term — finite, and a dilogarithm

$$\int dp \frac{p}{p+k^+} \left(\log \frac{\Xi}{p} + \frac{1}{2} \right) = p \left(\log \frac{\Xi}{p} + \frac{3}{2} \right) - k^+ \left[\left(\log \Xi + \frac{1}{2} \right) \log(p+k^+) - \log p \log \left(1 + \frac{p}{k^+} \right) - \text{Li}_2 \left(-\frac{p}{k^+} \right) \right]$$

$$\int dp \frac{p}{p + k^+} = p - k^+ \log(p + k^+)$$

(first primitive checked against quadrature to $3 \cdot 10^{-21}$). Both are finite at $p^+ \rightarrow 0$, so they contribute no divergence — only the term in §8.1 does.

9. The final result

$$\frac{d^3 N_{2\text{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \int_{\Lambda}^{\vee - k^+} \frac{dp^+}{2\pi} \int_{w,x,x',y'} \left\{ \mathcal{G}_{\text{fin}}(p^+) + \mathcal{G}_{\log}(p^+) \right\}$$

and the p^+ integral of the closed piece is

$$-\frac{g^4}{64\pi^7 k^+} \int_{w,x,x',y'} e^{-ik \cdot (y' - w)} \mathcal{C}_{\infty} \mathcal{K} \left[\log^2 \frac{\Xi}{\Lambda} - \log^2 \frac{\Xi}{\vee - k^+} \right] + (\text{finite, §8.2})$$

with

$$\begin{aligned} \mathcal{K} &= \frac{(x - w) \cdot (x' - y')}{2(x - w)^2 (x' - y')^2}, & \mathcal{K}_k &= \frac{[(x - w) \cdot \hat{k}][x' - y'] \cdot \hat{k}}{2(x - w)^2 (x' - y')^2}, \\ \Xi &= \frac{2e^{-\gamma_E} k^+}{|k_{\perp}| \rho_0}, & \mathcal{C}_{\infty} &= [U(y') - U(x')]^{c'e'} \rho^{e'}(x') \left\{ f^{c'ba} f^{cbd} U^{ac}(w) \rho^d(x) + N_c U^{c'e}(x) \rho^e(x) \right\} \\ \mathcal{G}_{\text{fin}}(p^+) &= \int d^2 u \left[\mathcal{I}(u) - \theta(|u| - \rho_0) \mathcal{I}_{\text{asy}}(u) \right] \quad \text{— absolutely convergent, needs the target.} \end{aligned}$$

10. Reading the answer

- **The whole Λ -dependence is now explicit**, and it is a *double* logarithm. That is the characteristic signature of a soft-collinear region, and it is why the $\left[\frac{1}{p^+}\right]_+$ prescription could not be applied.
- ρ_0 **cancels**. It appears in Ξ inside the $\log^2$ and, with the opposite sign, in $\mathcal{G}_{\text{fin}}$. Any convenient value works; the physics does not depend on it.
- **Nothing needed dimensional regularisation**. There was no UV and no collinear pole — the phase supplied the upper cutoff and ρ_0 is an artificial splitting point, not a regulator.
- **What remains** is $\mathcal{G}_{\text{fin}}$, which is a genuinely convergent five-fold transverse integral containing $U(z)$, to be done once the target is specified (MV model, or numerically on JIMWLK configurations).
- **Before you use any of this**, apply the fixes from the previous note: the $(z - w)^{k'} \rightarrow (x - w)^{k'}$ typo in the $-\frac{S^2}{p^{\tau/2}}$ group, the uniform Wilson-line index ordering, and the measure $(\int_y \text{out}, \int_{x'} \text{in})$.

Checked by `src/row27_asy.py`, `row27_fix.py`, `row27_u.py`, `row27_final.py`; output in `RESULTS_row27.txt`.